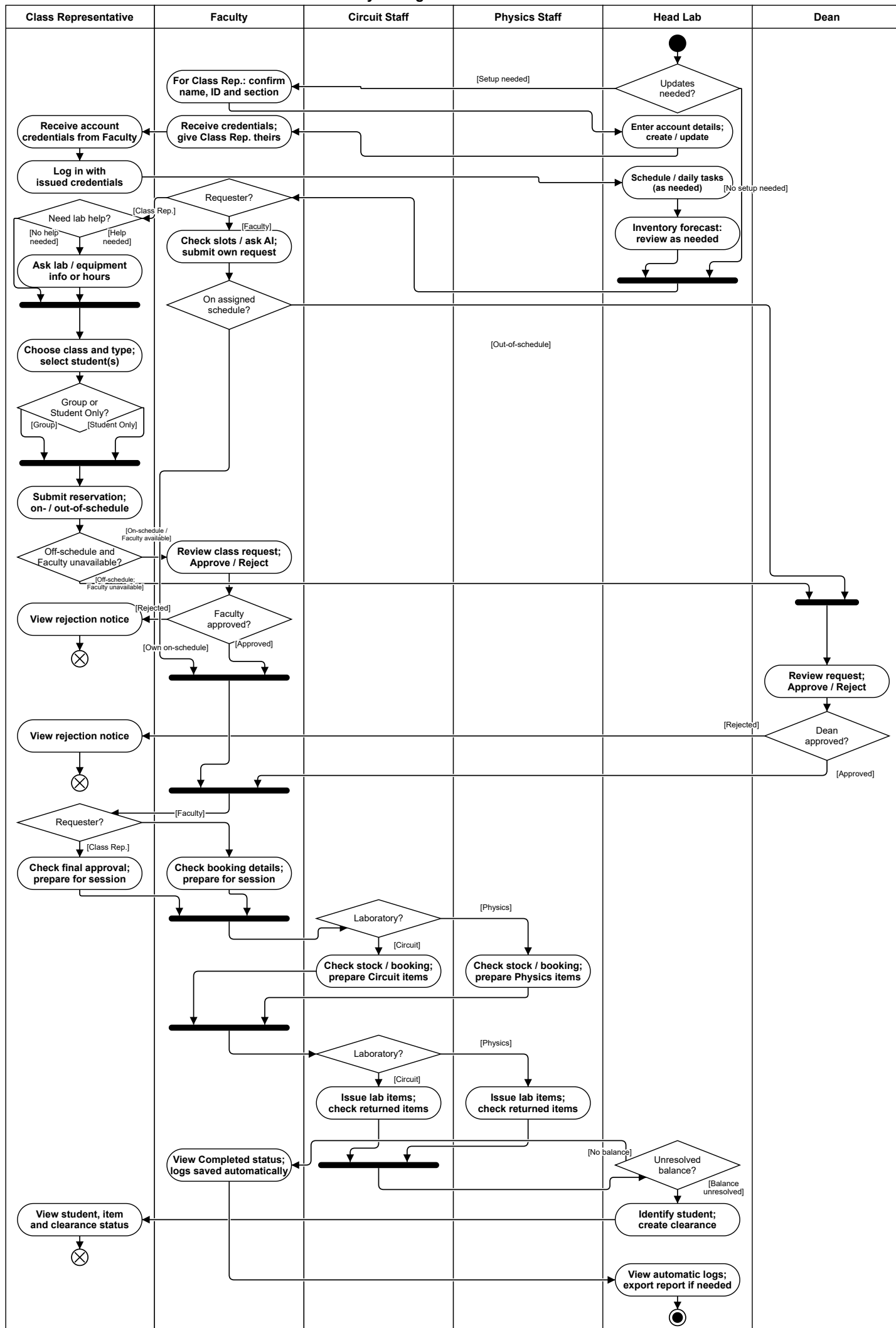
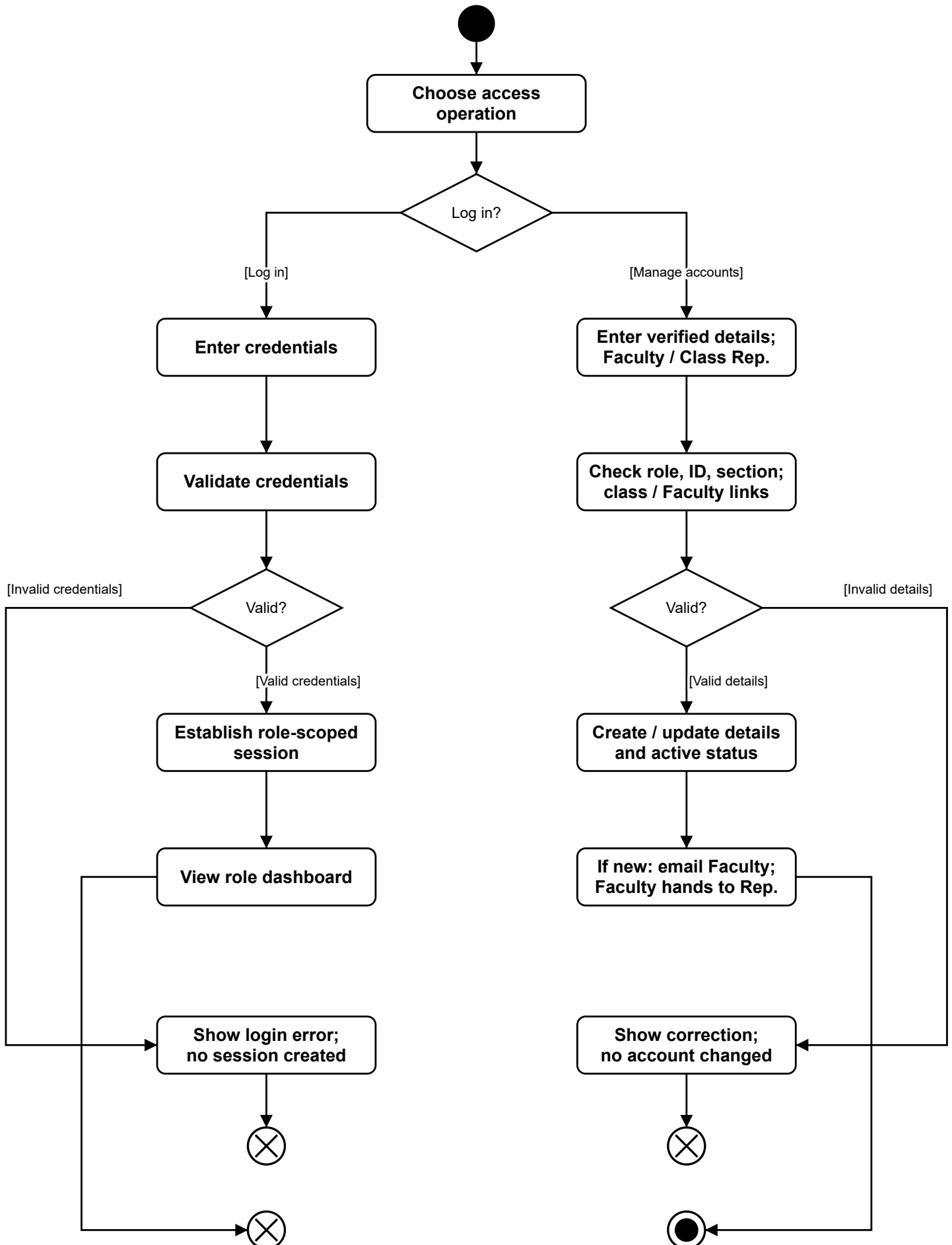


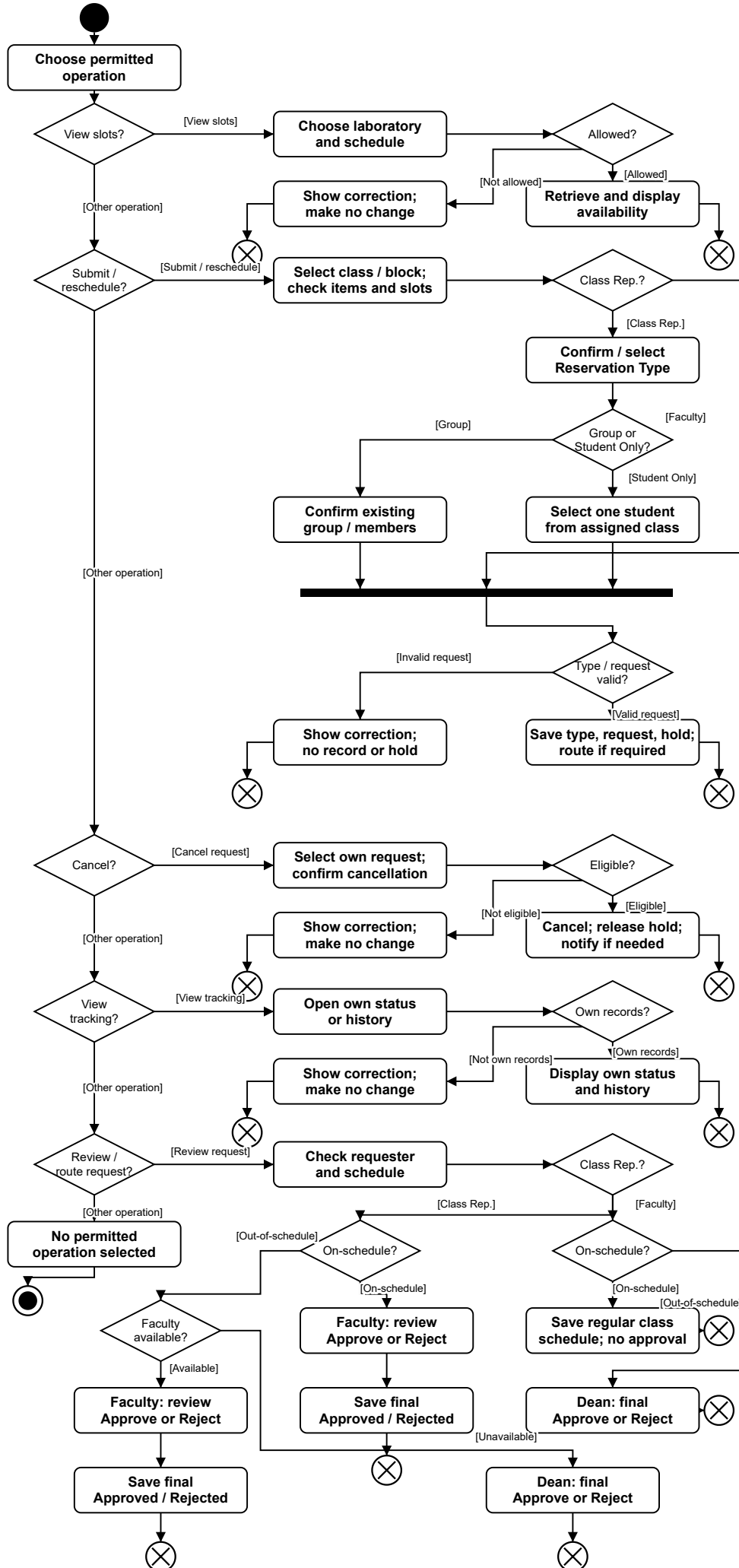
Laboratory Management — Overall Swimlane



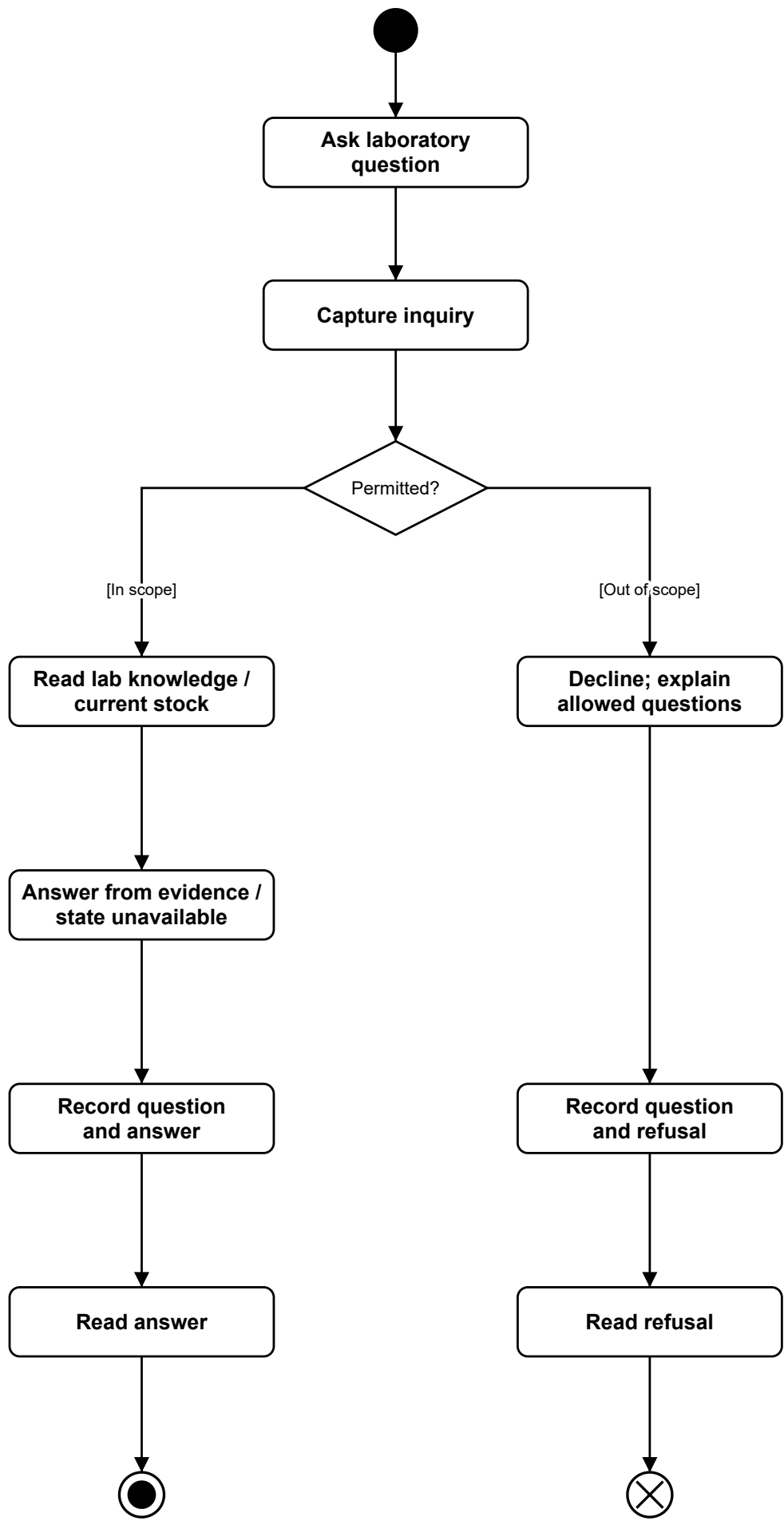
Login: any role. Create / update Faculty or Class Rep. accounts: signed-in Head Laboratory only.



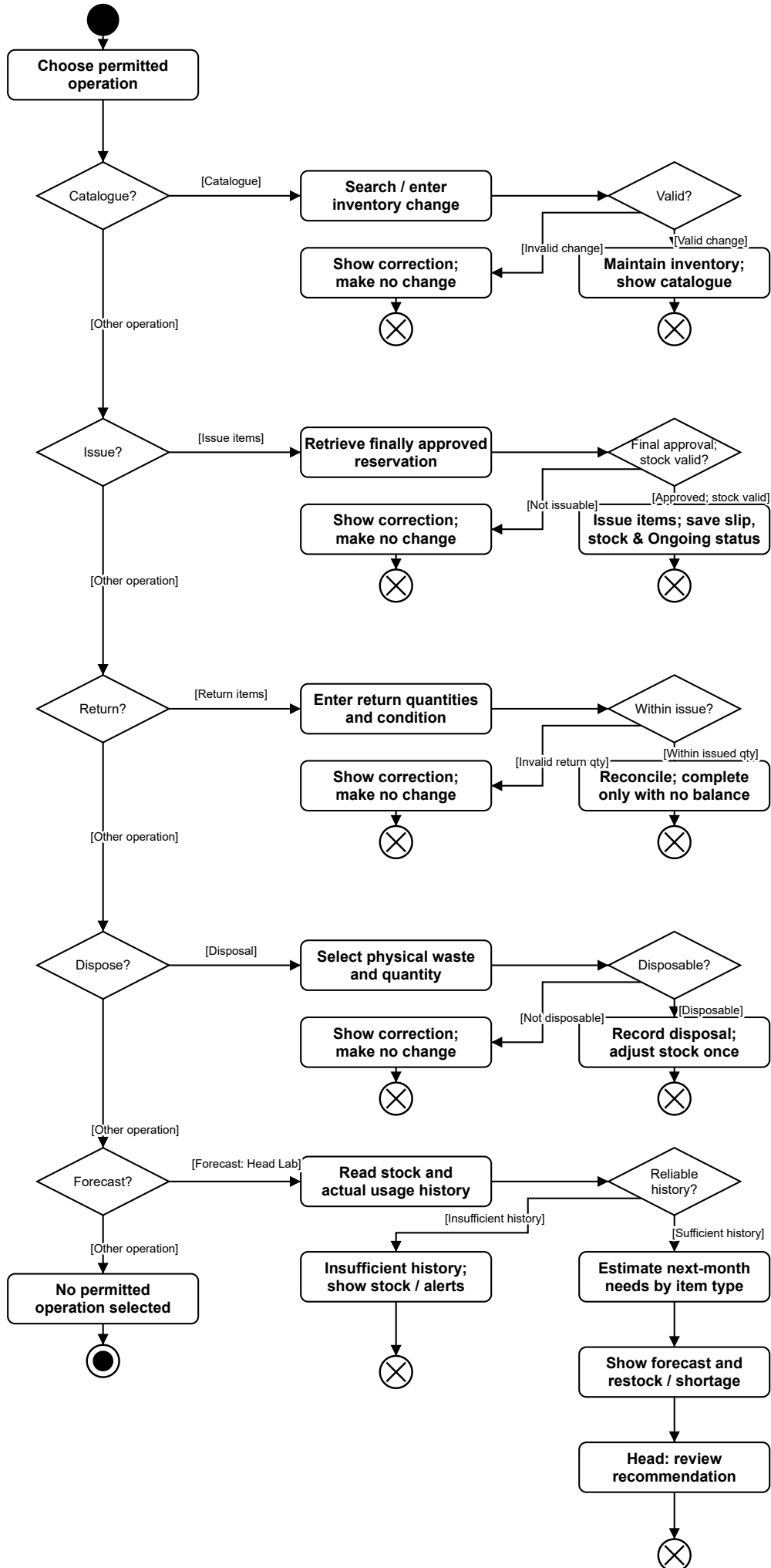
Class Rep.: choose Group or Student Only for BOTH schedule variants. Student Only = one selected class student.
On-schedule → Faculty; out-of-schedule → available Faculty, or Dean if Faculty is unavailable.



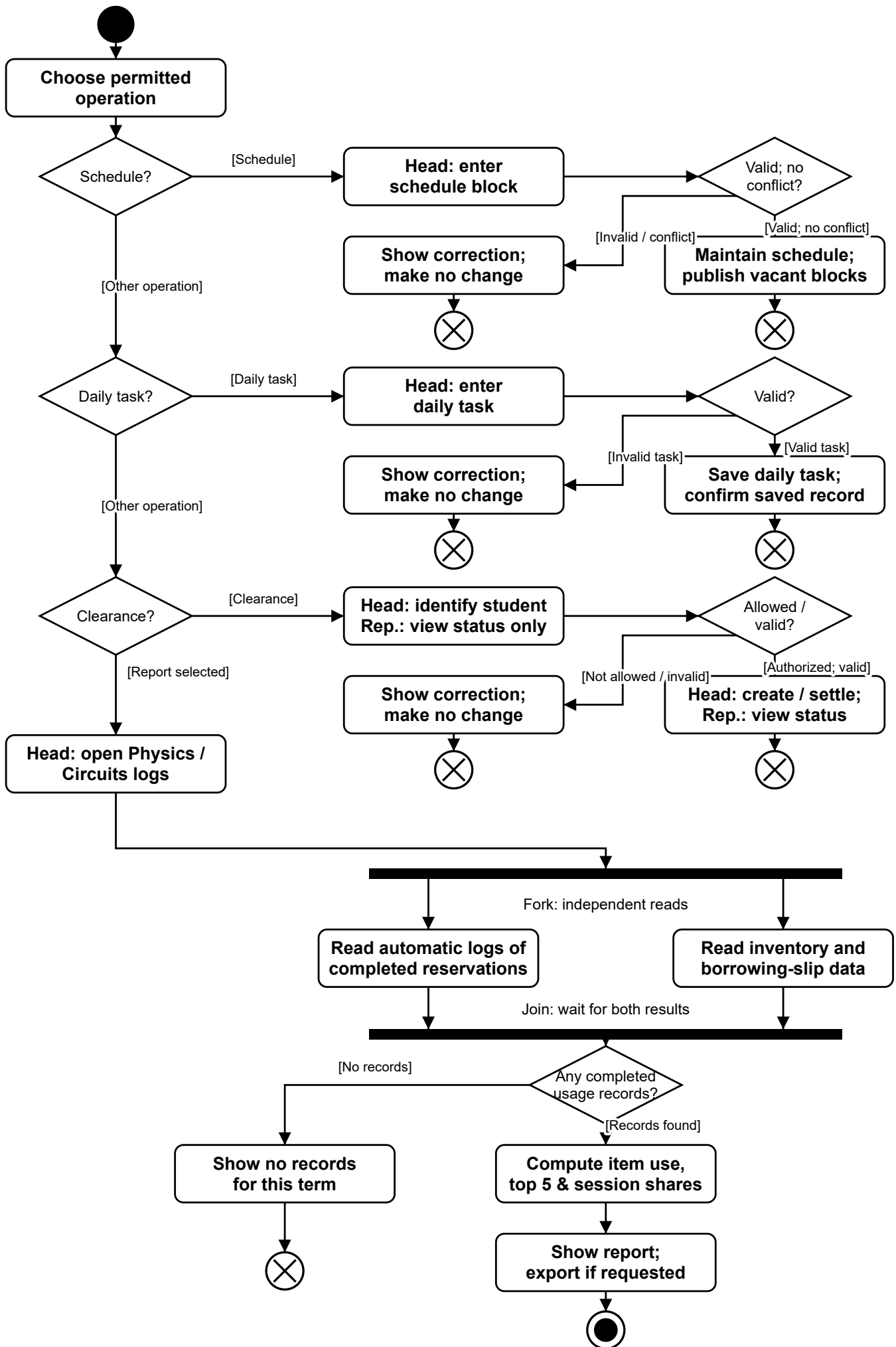
Precondition: signed-in Class Representative or Faculty; informational Q&A only.



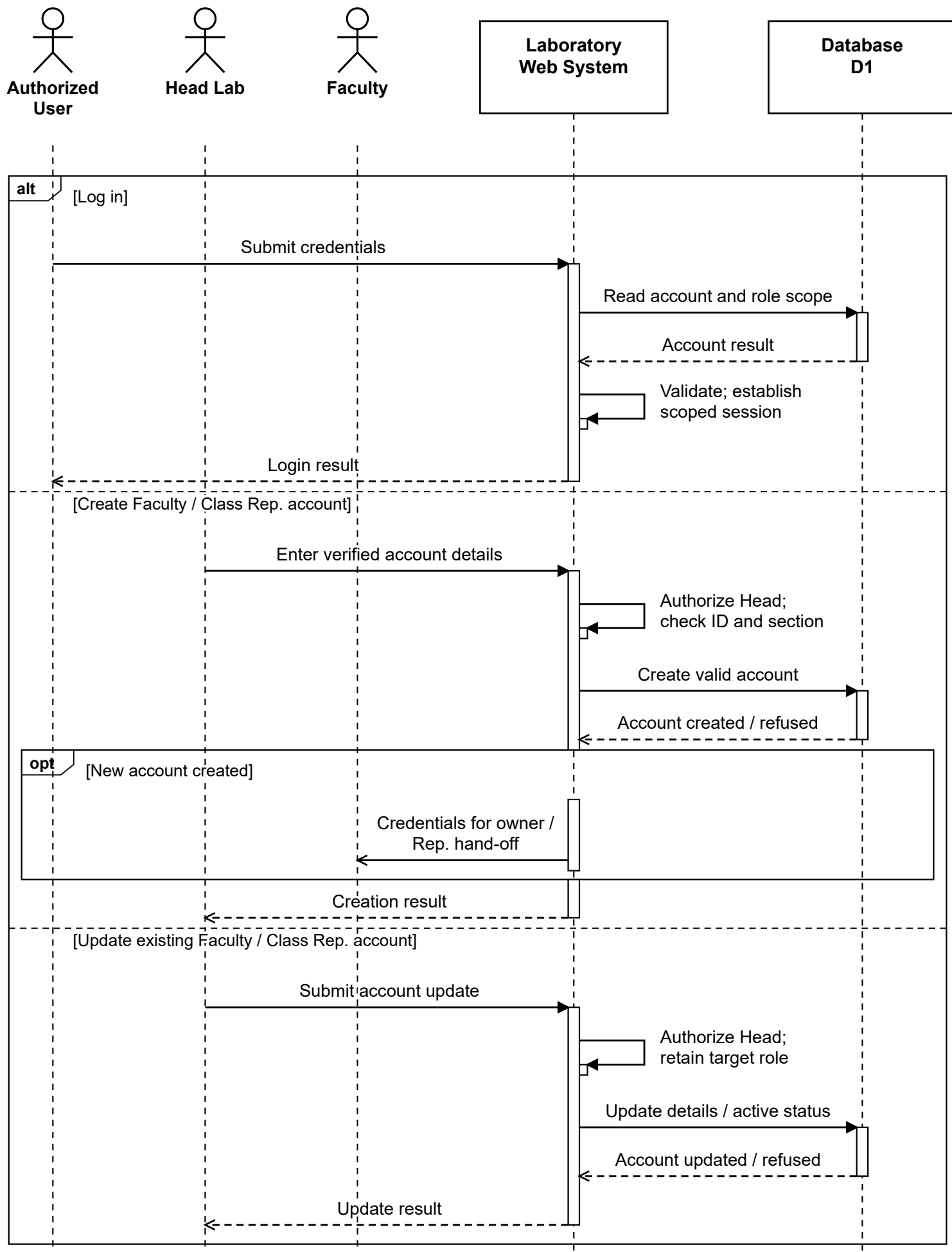
Precondition: signed-in Head Laboratory or Staff within the assigned laboratory.



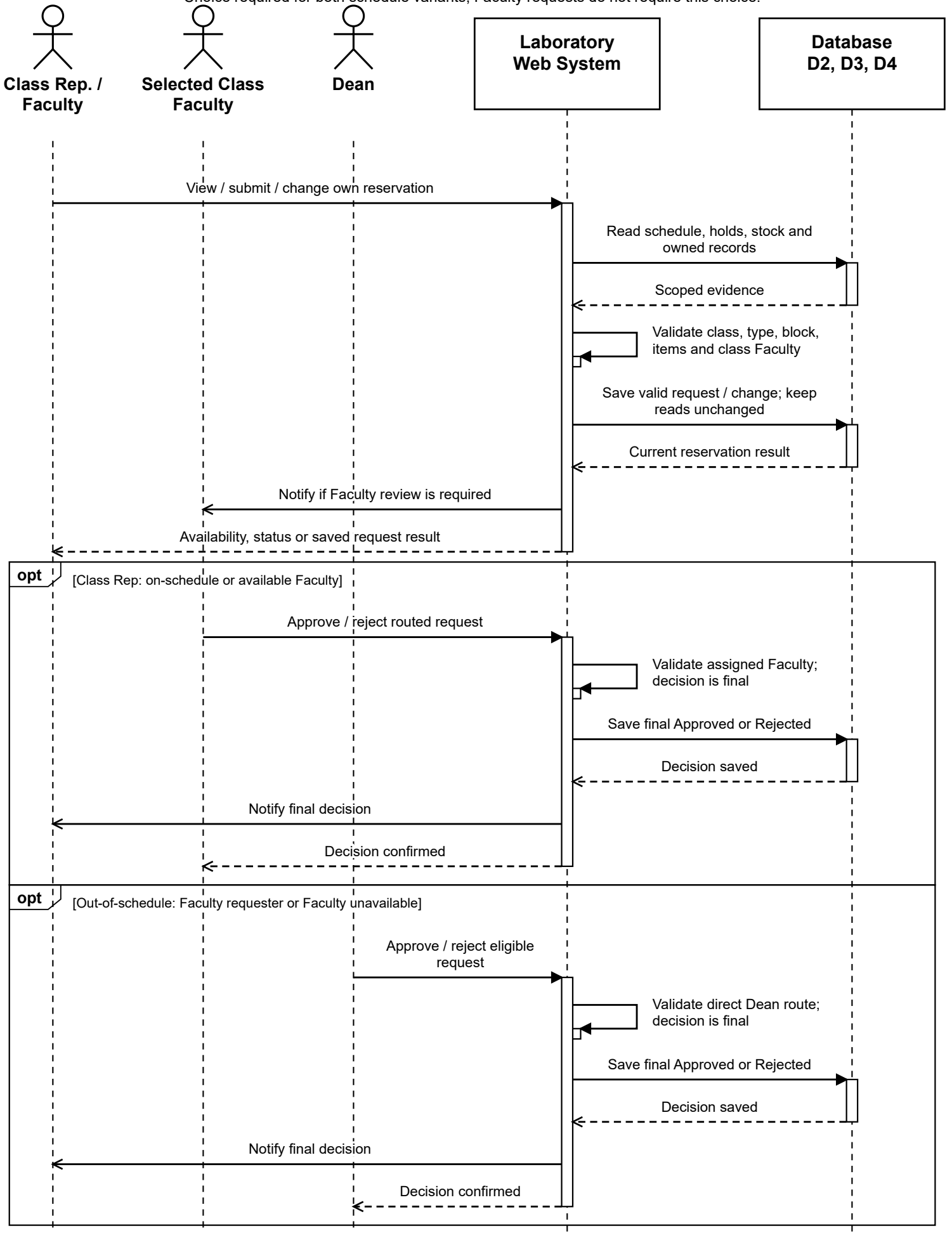
Precondition: signed-in Head Laboratory; Class Representative: assigned-class student clearance only.



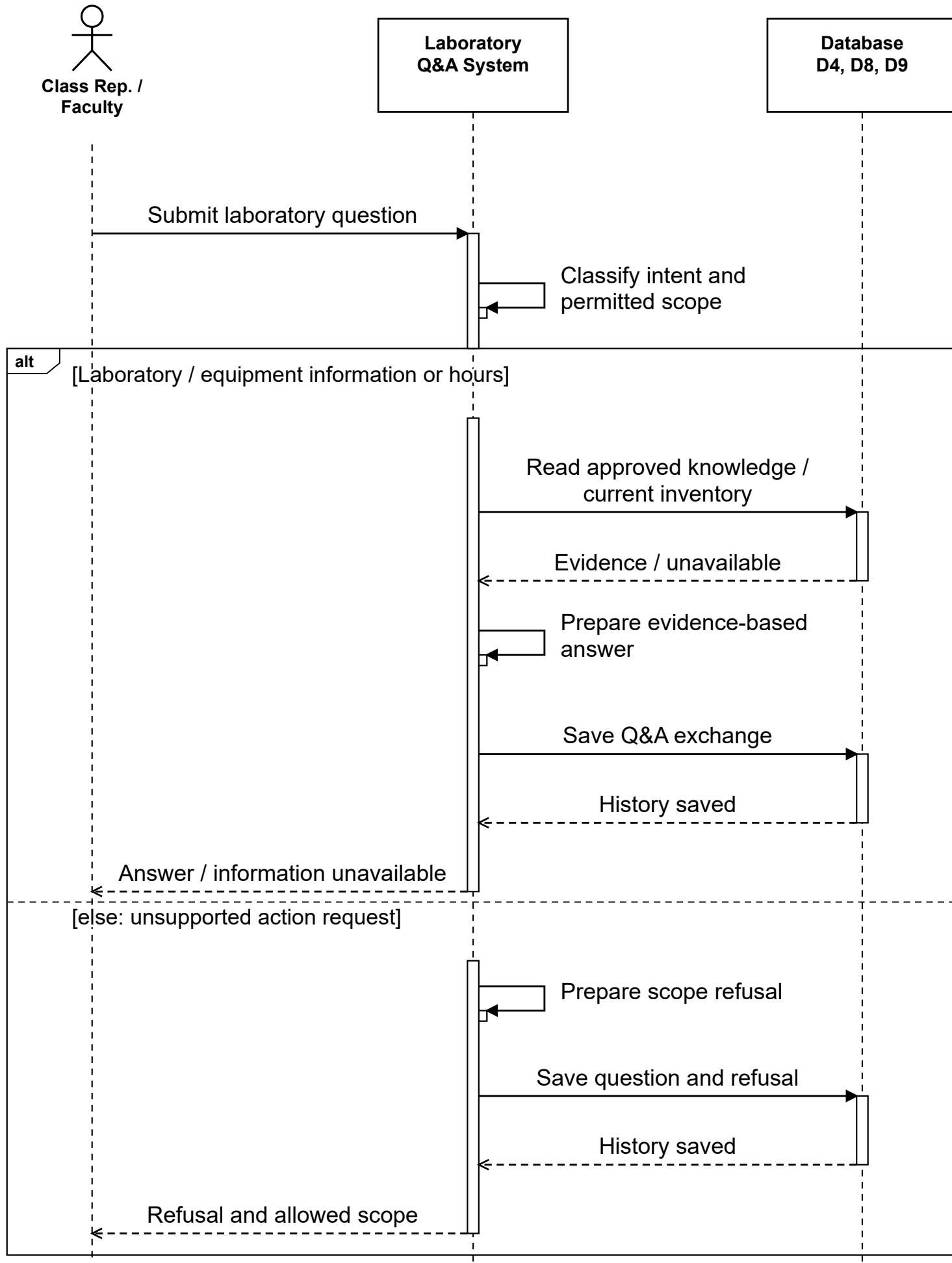
Login uses an issued account. Head creates Faculty first; Faculty manually verifies Class Rep. name, Student ID and section.



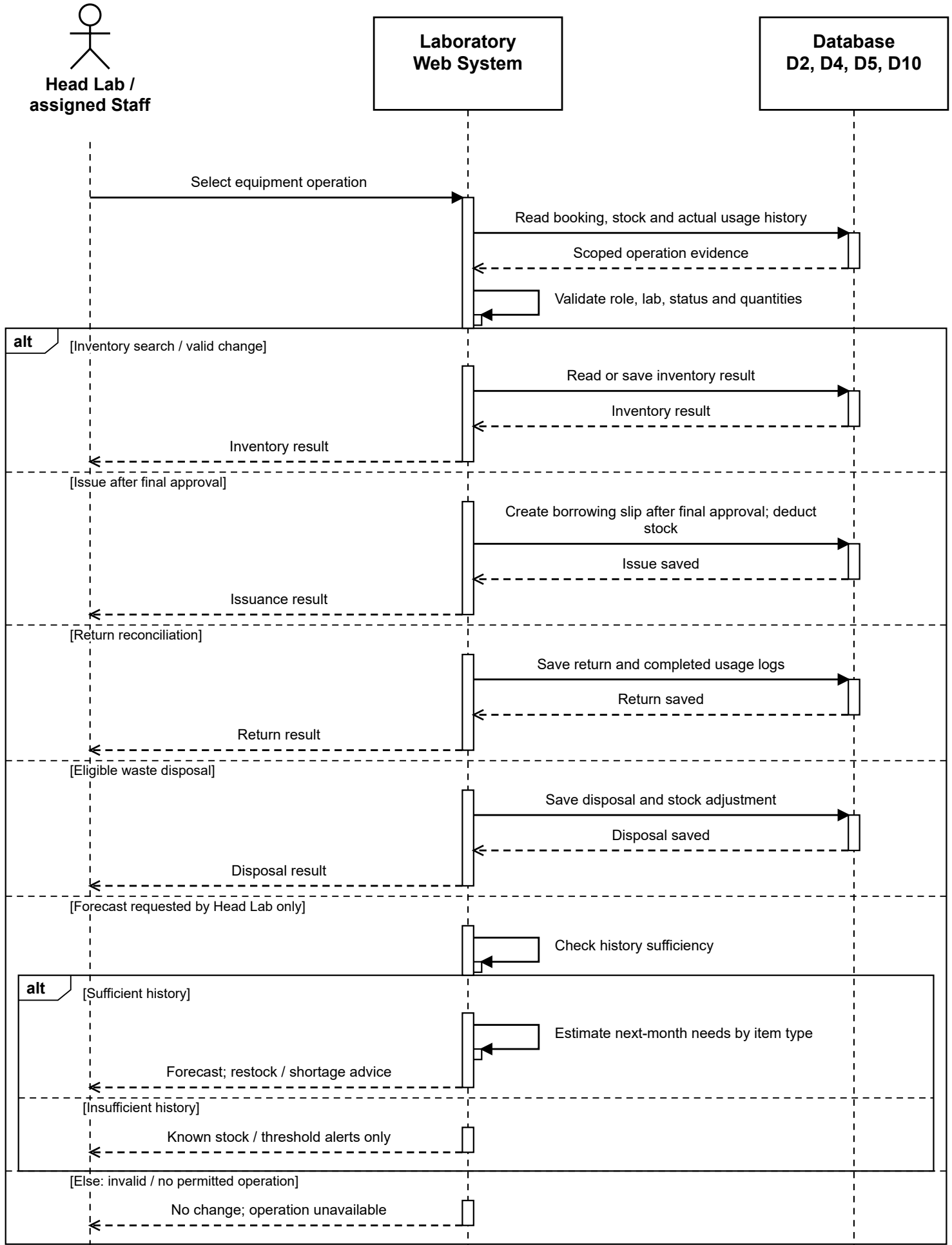
Class Rep.: Group = selected class members; Student Only = one selected class student.
Choice required for both schedule variants; Faculty requests do not require this choice.



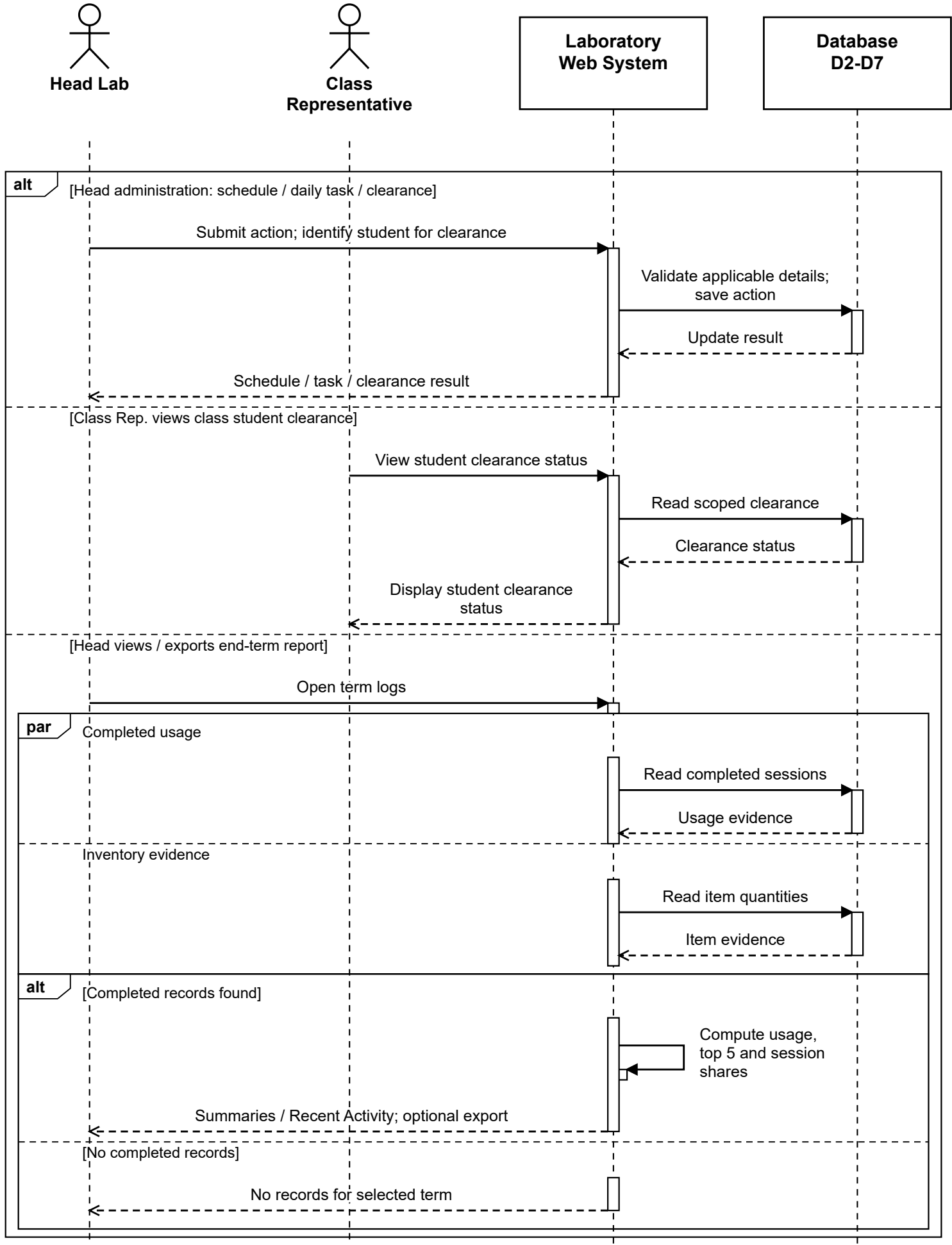
Signed in Class Representative or Faculty; Q&A is informational and scope-limited.



Signed in Head Laboratory or assigned laboratory Staff; laboratory scope is enforced.

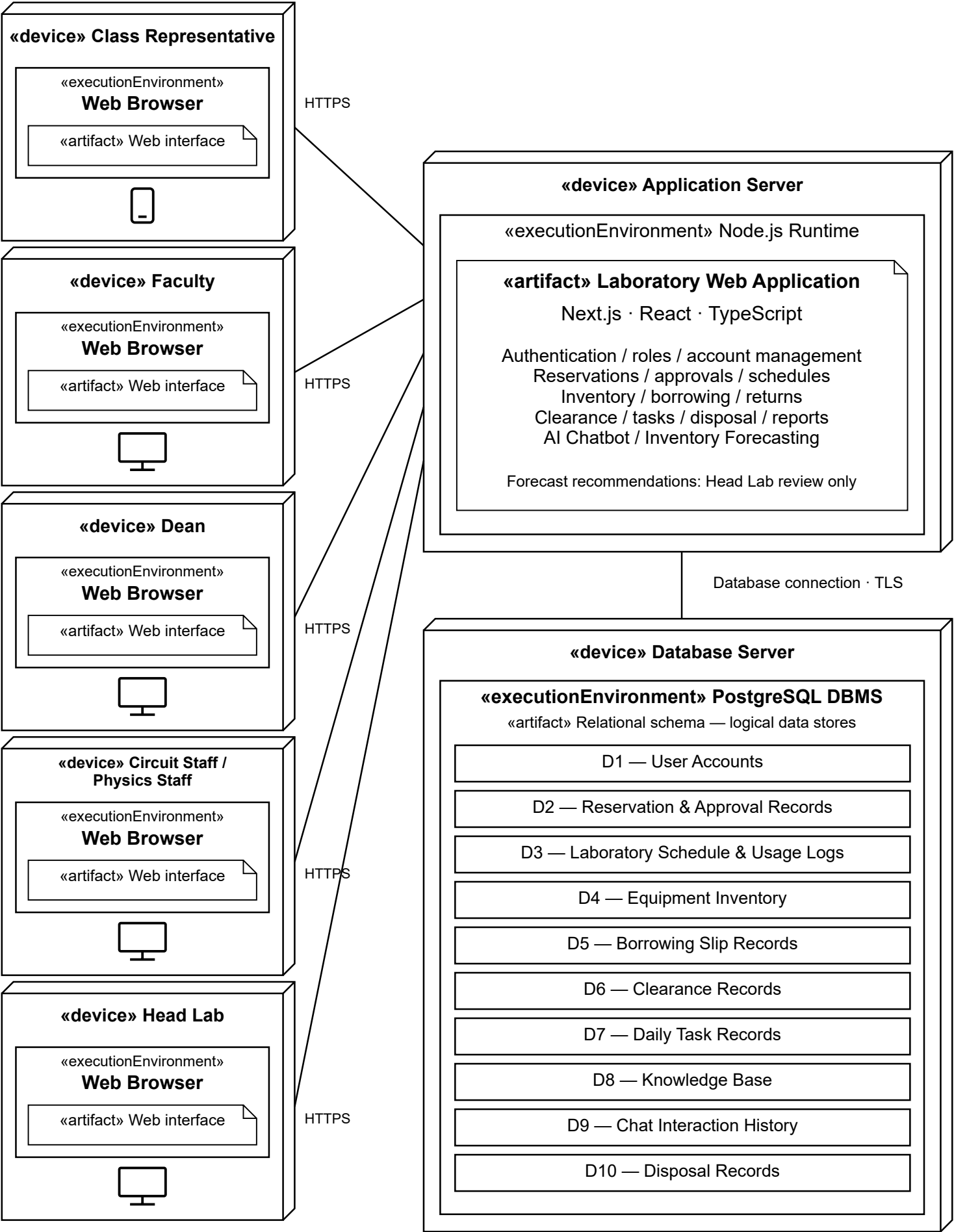


Head identifies the responsible student before creating clearance. Class Rep only views own-class clearance status.



Physics and Circuits Laboratory Management System

Deployment Diagram · Proposed architecture



All roles: Desktop / Laptop / Tablet / Smartphone · Chrome / Edge / Safari

One application and one database · No direct client database access

AI runs in the backend; no external provider selected · Forecasts never automatically change stock